GEMM

Team 8

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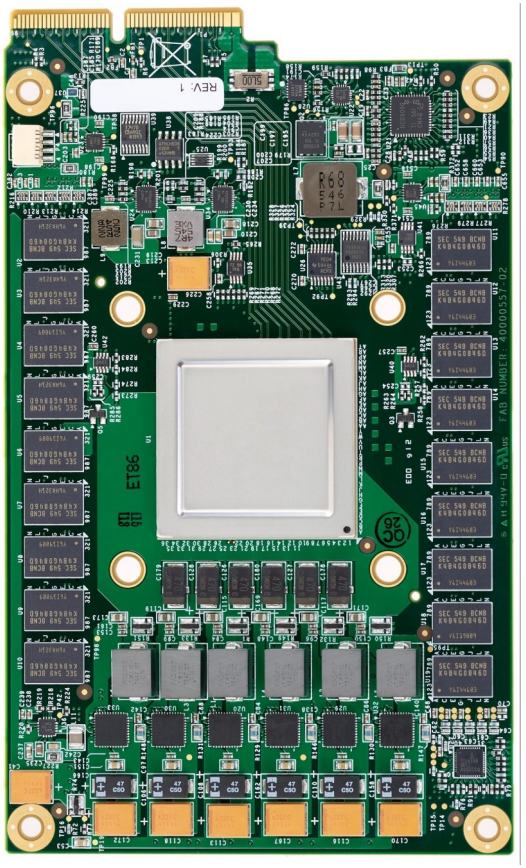
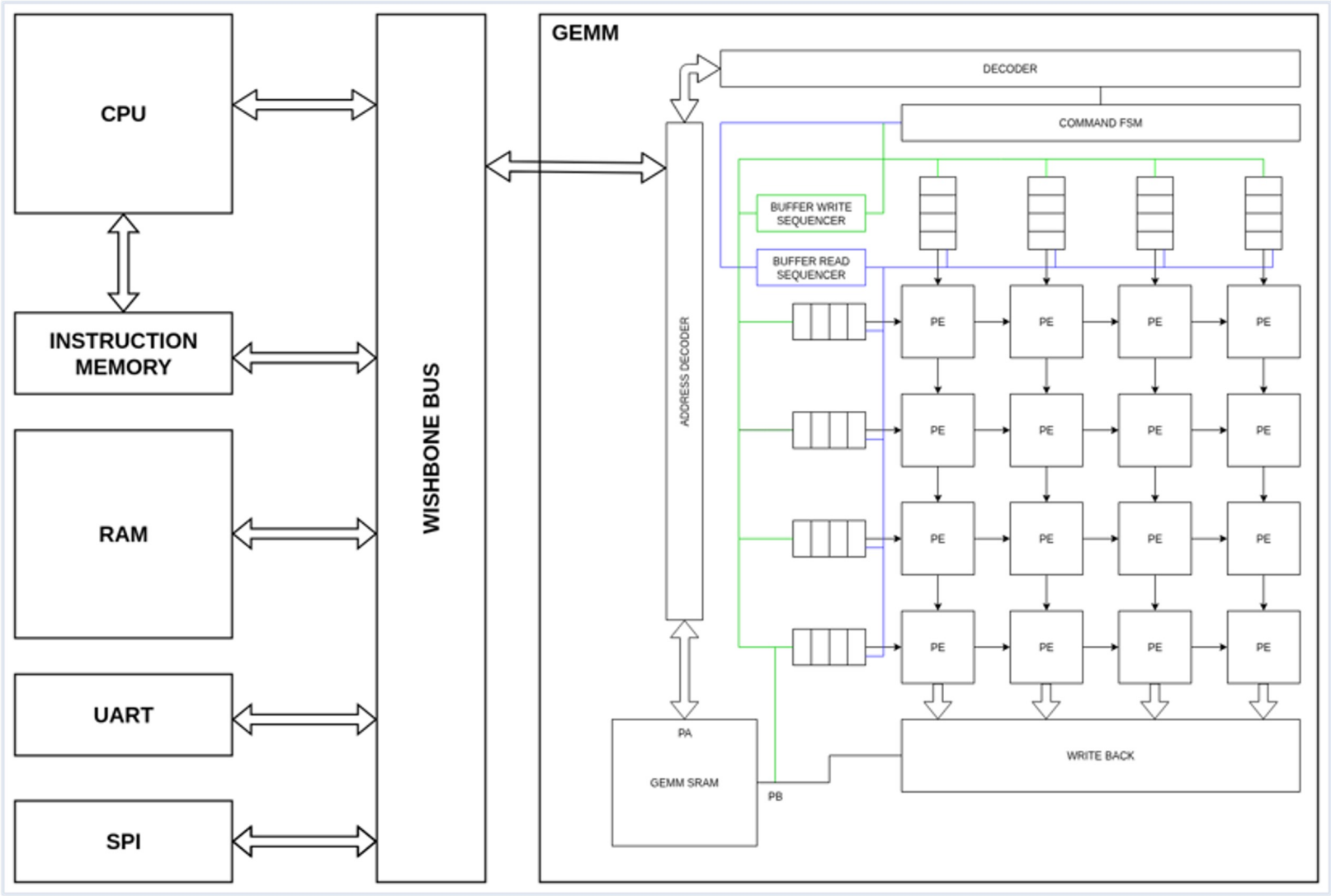
18/06/2026

Background & Motivation

- GEneral Matrix to Matrix multiplication is the core kernel behind machine learning, computer graphics, and signal processing
- Defined as C = αAB+βC
- We perform GEMM in silicon using a 4 by 4 systolic array
- Systolic arrays allow us to keep data flowing locally, maximize operand reuse, and packs parallelism into a small, layout-friendly structure
- To support matrices larger than the hardware array dimensions, tiling and workload orchestration are handled in C firmware running on a RISC-V processor,



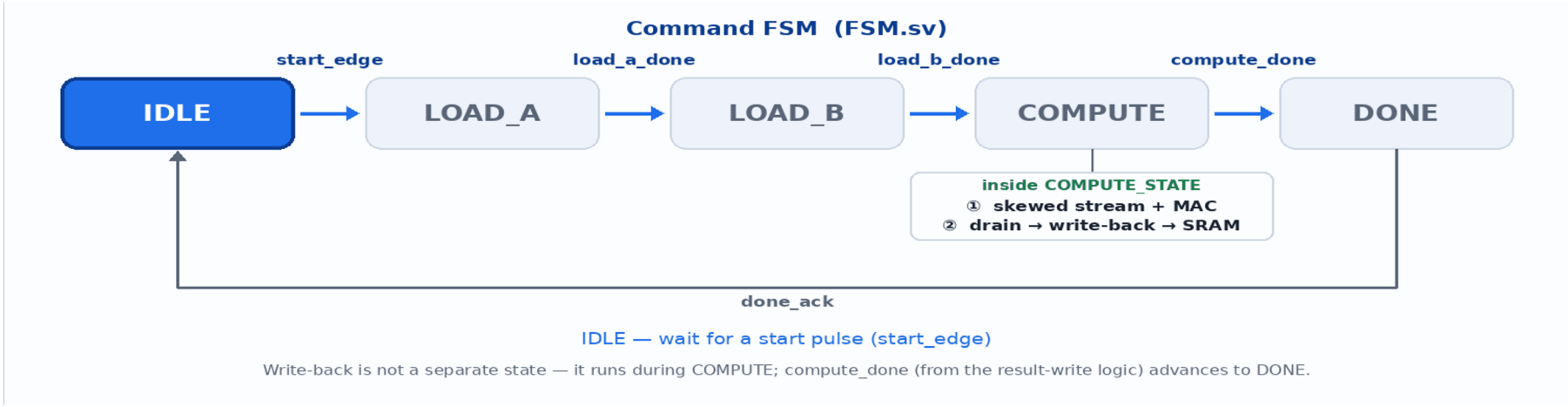
Google TPUv1 (2015) features a 256x256 systolic array of 8-bit integer multiply-accumulate units, delivering 92 teraops/s


CPU
INSTRUCTION MEMORY
RAM
UART
SPI
WISHBONE BUS
GEMM
DECODER
COMMAND FSM
BUFFER WRITE SEQUENCER
BUFFER READ SEQUENCER
ADDRESS DECODER
PE
PA
GEMM SRAM
PB
WRITE BACK

ISA + Memory Map

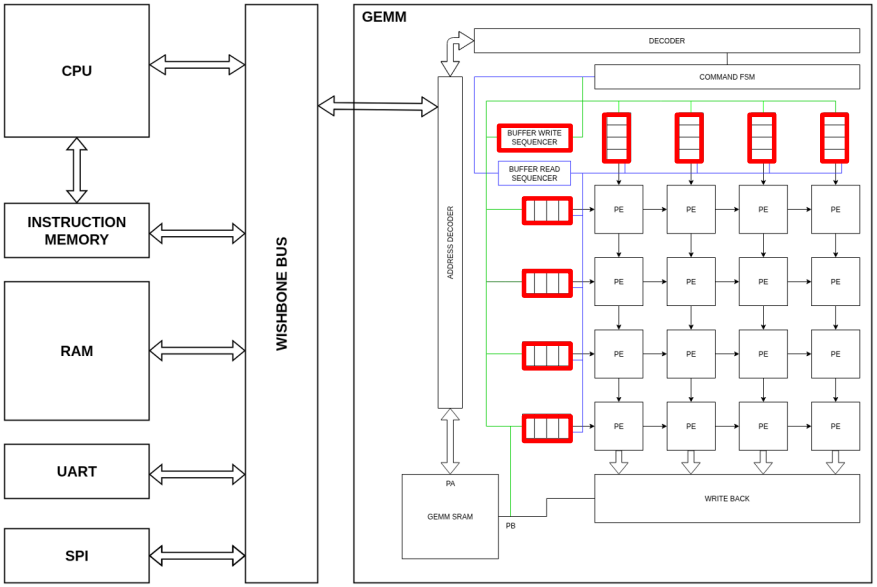
Instruction	Function
`MMINT8 a b c`	Matrix multiply; a/b/c = west / north / result base addresses
`TILE_BEGIN`	Preserve systolic array contents (start of tile)
`TILE_END`	Clear systolic array contents (end of tile)

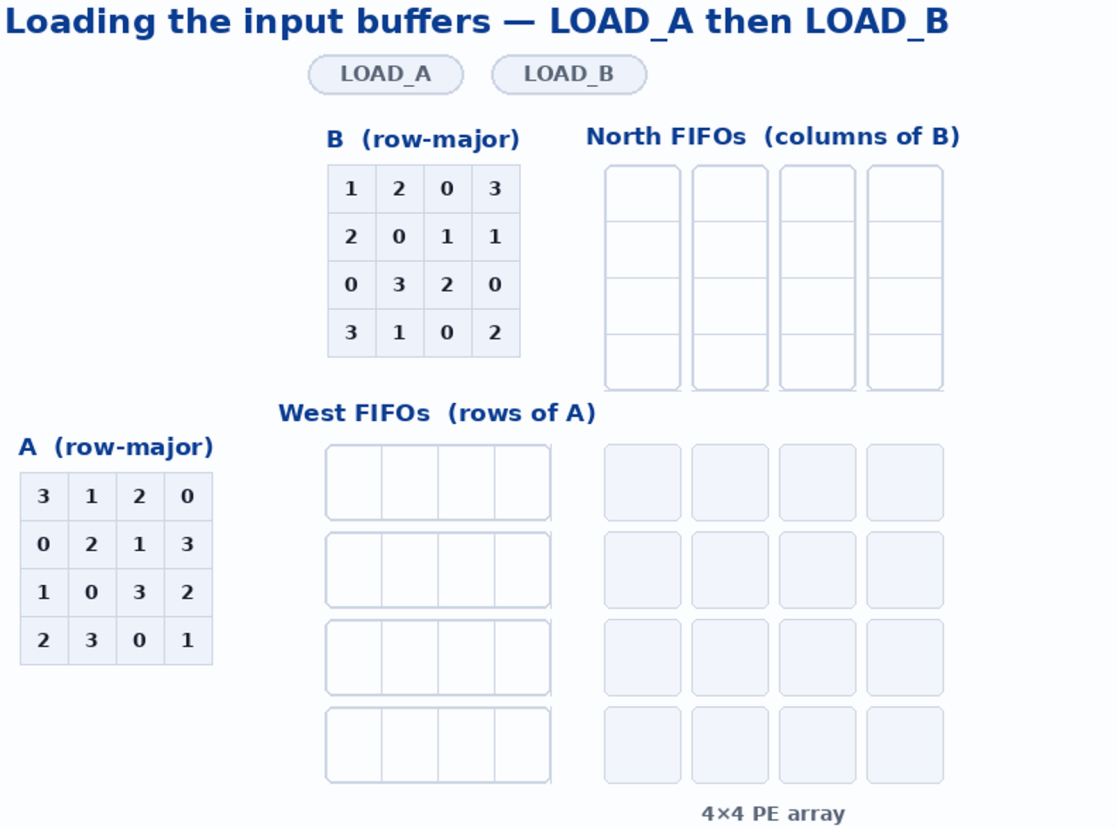
Offset	Register	Access	Purpose
`+0x0`	`INSTRUCTION_REG`	R/W	Issue instructions
`+0x4`	`STATUS_REG`	R	Computing / idle
`+0x8–0x87`	SRAM window	R/W	32 × 32-bit words (128 B)



LOAD

- Triggered by the command FSM
- Buffer Write Sequencer (BWS) streams operands out of the SRAM and into FIFOs
- West side buffers feed matrix A row-by-row, North side feed B across all 4 columns at once.
- Each is a 4-deep x 8-bit FIFO

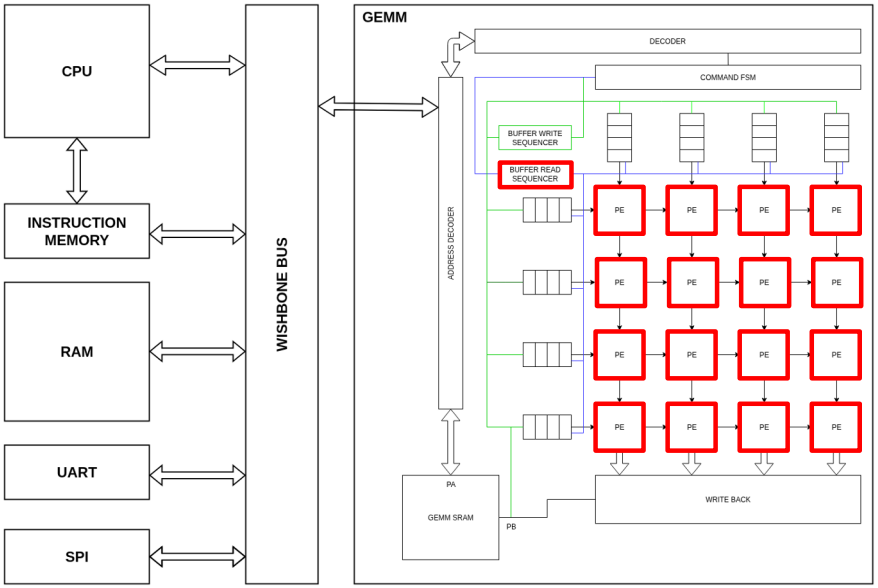


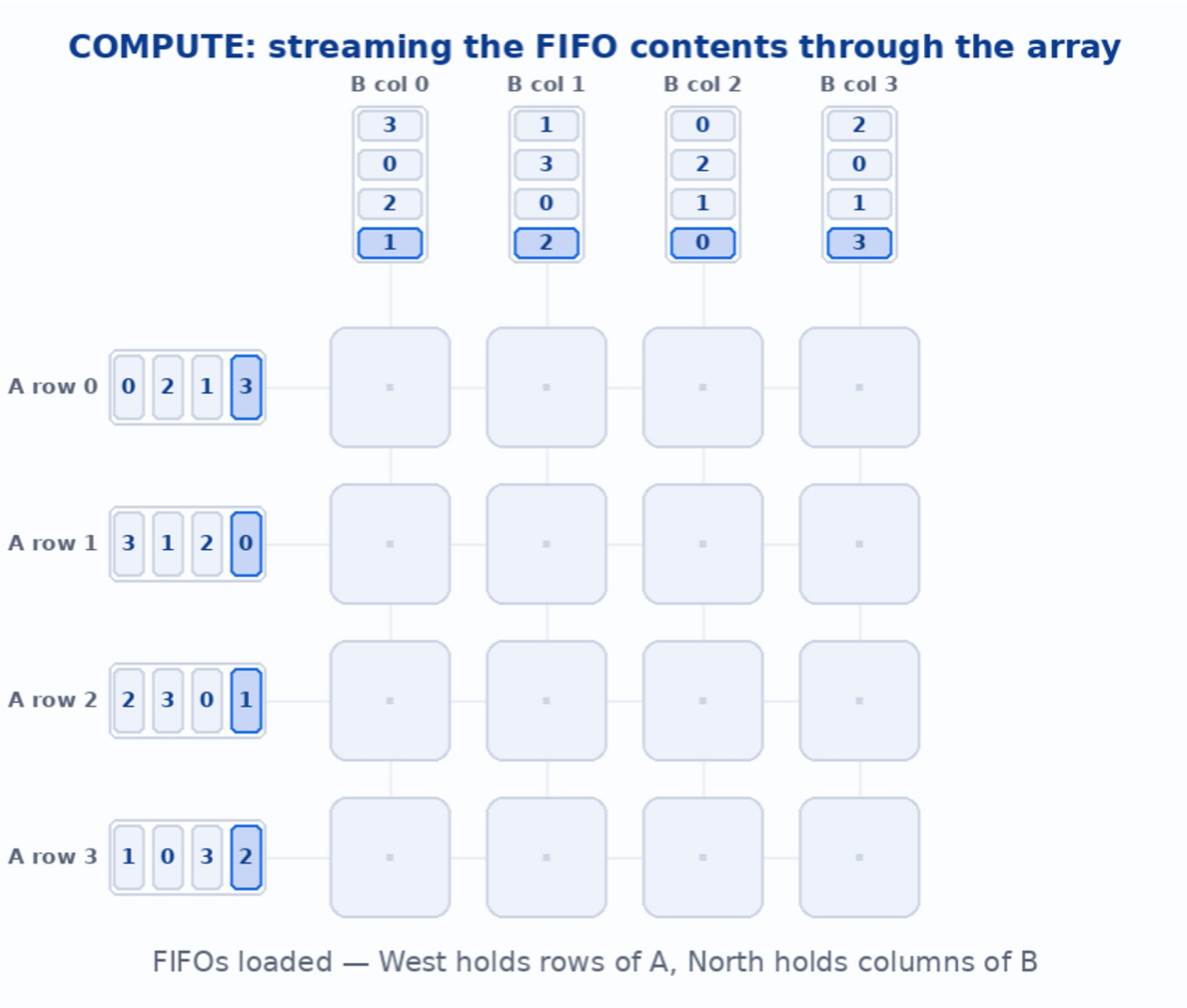


Operands sit in SRAM row-major → loaded into the West (A) and North (B) FIFOs

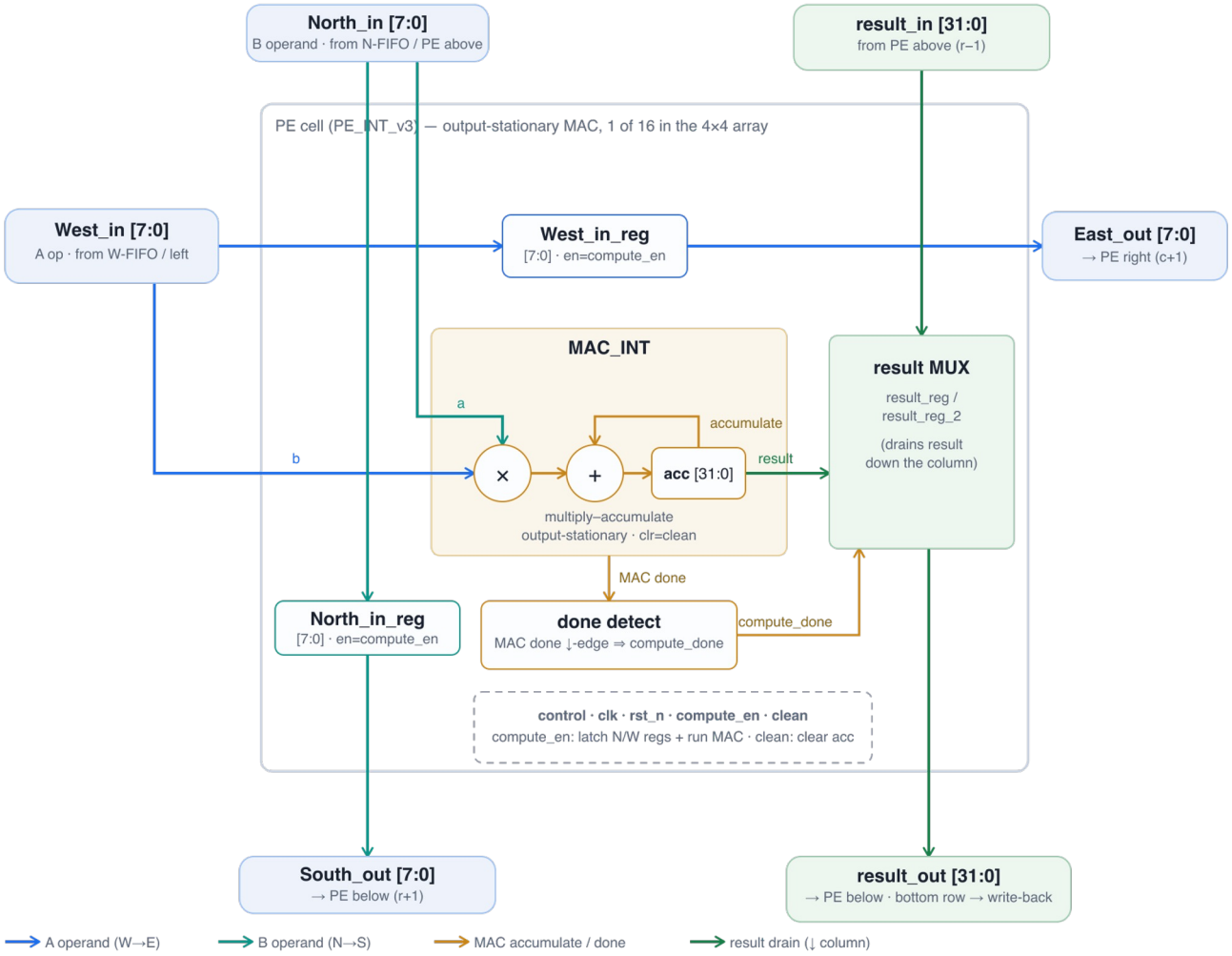
COMPUTE

- Buffer Read Sequencer (BRS) drains the input FIFOs in a skewed wavefront.
- Each PE holds a MAC unit - a signed 8 by 8 multiply accumulated into a 32 bit register.
- Valid bits propagate with the data so that PE only start multiplying once real operands reach it





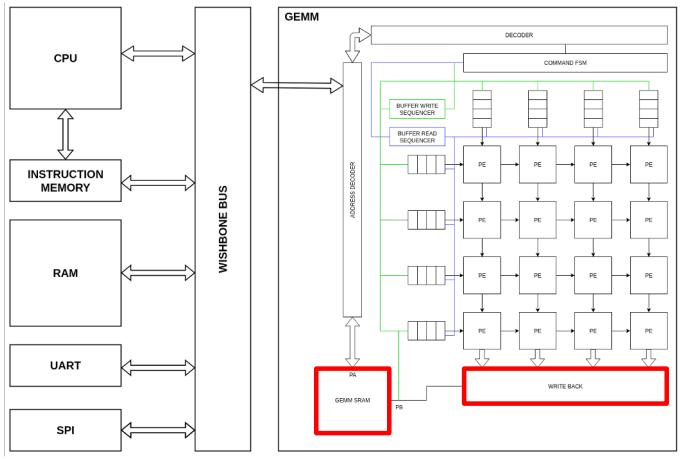
PE design

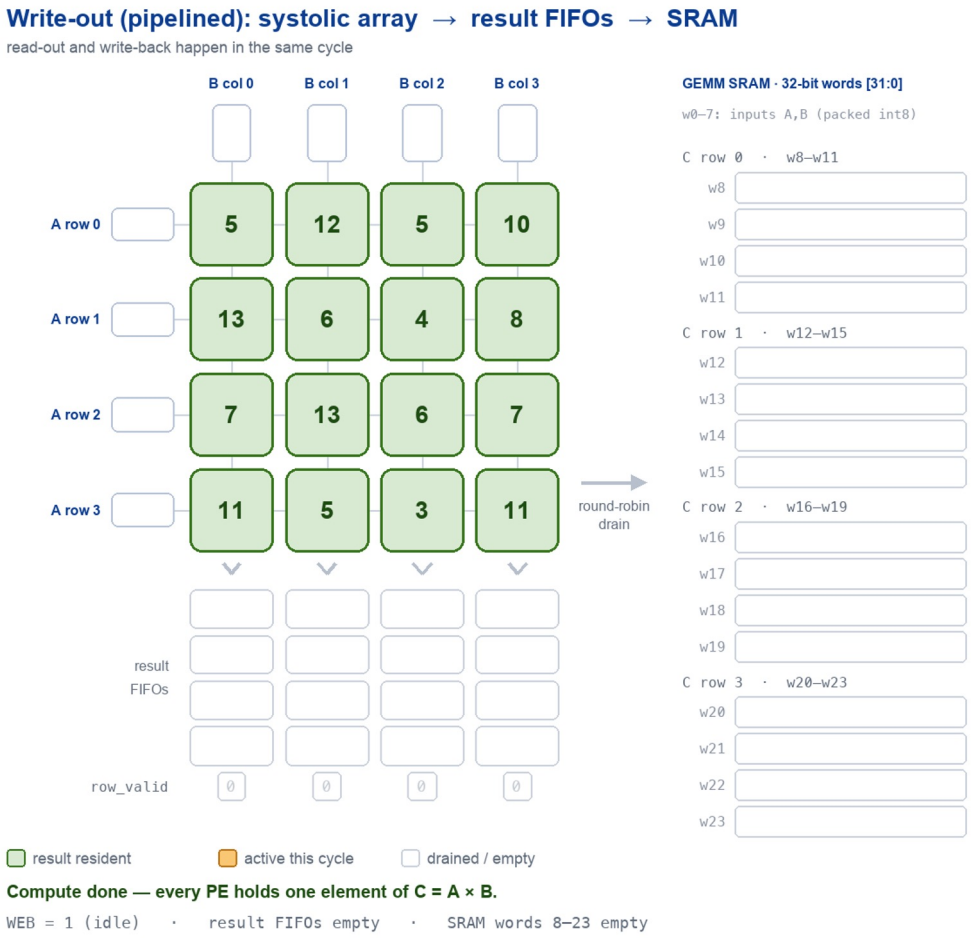

North_in [7:0]
B operand · from N-FIFO / PE above
result_in [31:0]
from PE above (r−1)
PE cell (PE_INT_v3) — output-stationary MAC, 1 of 16 in the 4×4 array
West_in [7:0]
A op · from W-FIFO / left
West_in_reg
[7:0] · en=compute_en
East_out [7:0]
→ PE right (c+1)
MAC_INT
a
b
accumulate
×
+
acc [31:0]
result
multiply–accumulate
output-stationary · clr=clean
result MUX
result_reg /
result_reg_2
(drains result
down the column)
MAC done
North_in_reg
[7:0] · en=compute_en
done detect
MAC done ↓-edge ⇒ compute_done
compute_done
control · clk · rst_n · compute_en · clean
compute_en: latch N/W regs + run MAC · clean: clear acc
South_out [7:0]
→ PE below (r+1)
result_out [31:0]
→ PE below · bottom row → write-back
A operand (W→E)
B operand (N→S)
MAC accumulate / done
result drain (↓ column)

WRITE-BACK

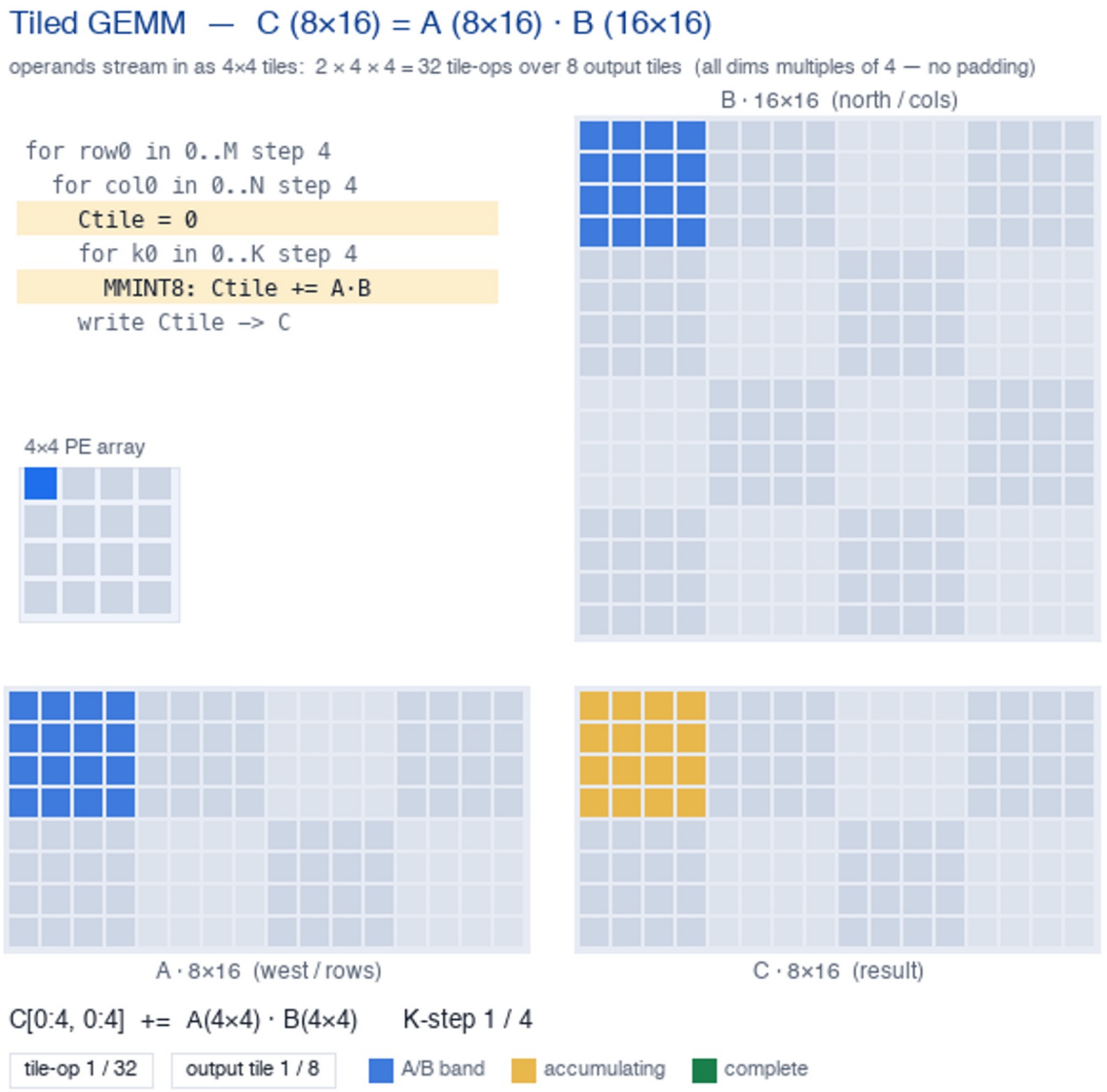
fully pipelined, zero-stall result drain

- One FIFO per column catches the array's anti-diagonal output (≤ 4 results/cycle)
- FIFOs decouple array ↔️ SRAM (1 × 32-bit word/cycle) → PEs never stall
- Read-out and write-back fully overlap — pipelined, not two phases
- Round-robin drain rebuilds C in row-major order, for free (no transpose)
- DONE auto-detected when FIFOs drain → CPU polls status, reads SRAM (no IRQ/DMA)





Software Tiling



BSP / HAL API — UART & GEMM Accelerator

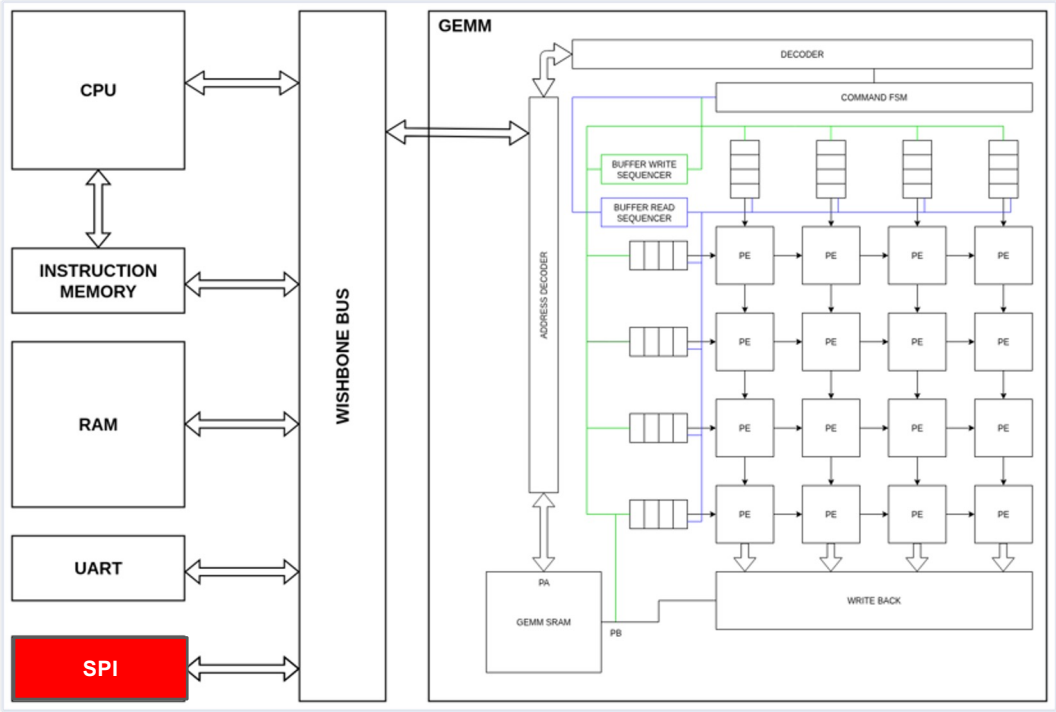
Board-support functions for driving the on-chip UART and the GEMM matrix-multiply IP (mono = function, right = behaviour)

UART HAL	
`uart_putc(int c)`	Transmit one byte; blocks while the UART TX is busy
`uart_getc(void)`	Receive one byte; blocks until RX is ready (returns the byte)
`uart_send_u32(int v)`	Transmit a 32-bit word as 4 bytes, little-endian (LSB first)
`uart_recv_u32(void)`	Receive a 32-bit word from 4 bytes, little-endian (LSB first)
GEMM HAL — high-level	
`gemm_init(void)`	Reset / arm the accelerator before a run
`gemm_run(A, a_words, B, b_words, C, c_words, V)`	One 4×4 tile, end-to-end: load A,B → MMINT8 → wait → read C
`gemm_run_auto_tile_int8(A, U, V, B, W, C)`	Arbitrary-size INT8→INT32 GEMM: auto 4×4 tiling + K-accumulation
GEMM HAL — control & status	
`gemm_issue_tile_begin(void)`	Issue TILE_BEGIN — open a tile, preserve the PE accumulators
`gemm_issue_mmint8(ly)`	Issue MMINT8 a b c — start a 4×4 matmul from the SRAM operands
`gemm_issue_tile_end(void)`	Issue TILE_END — acknowledge DONE and clear the array
`gemm_wait_done(void)`	Block (spin) until the STATUS register reads DONE
`gemm_status() / gemm_is_done() / gemm_is_busy()`	Read the command-FSM state and derive done / busy flags
GEMM HAL — operand / result memory	
`gemm_load_A_words(ly,…) / gemm_load_B_words(ly,…)`	Write packed A / B operands into the GEMM SRAM
`gemm_read_C_words(ly,…)`	Read the INT32 result tile back from the GEMM SRAM
`gemm_sram_write_words() / gemm_sram_read_words()`	Raw word-granular GEMM-SRAM read / write

Internal helpers used by gemm_run_auto_tile_int8: gemm_layout · pack_int8_to_words · extract_A/B_tile_int8 · scatter_C_tile_int32 · gemm_pack_instr / gemm_make_*_instr · run_one_tile

Why SPI?

For peripheral communication, debug, and demonstrating a complete memory-mapped control-oriented IP block.



SPI microarchitecture

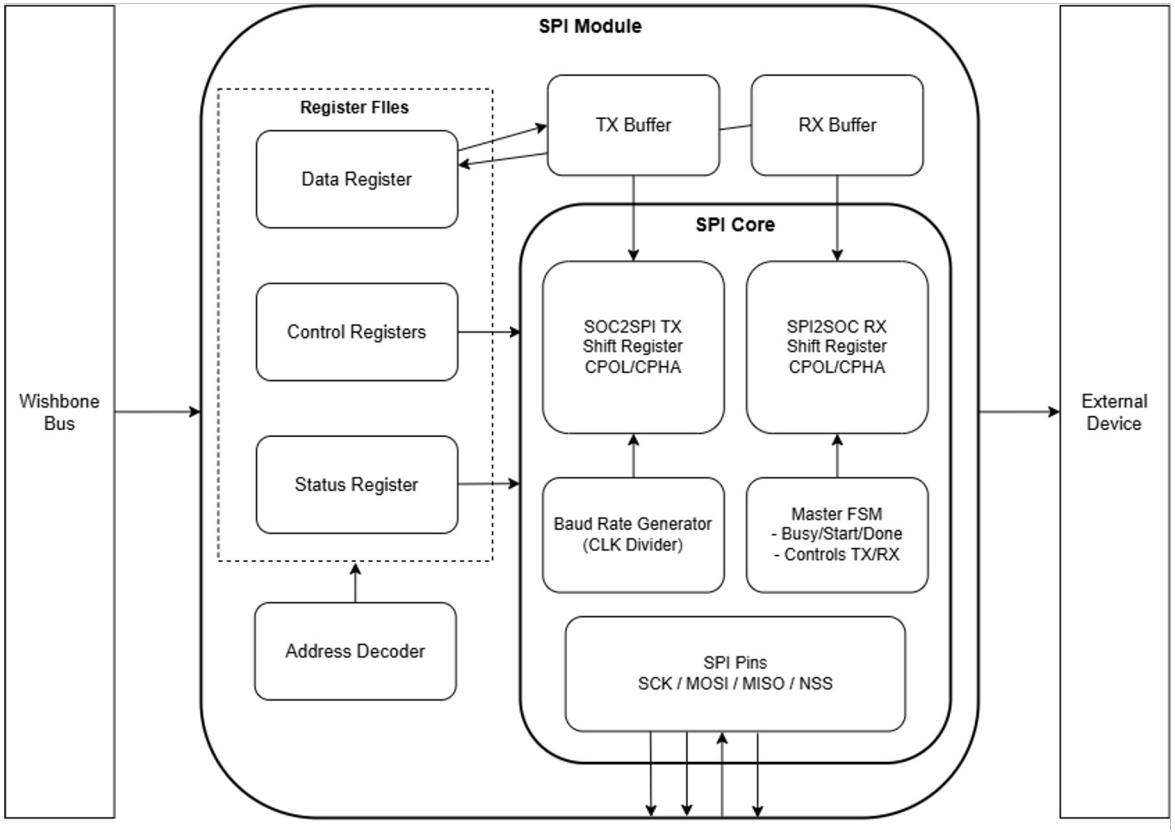
Dual Path Design

Bus Mode:

Traditional SPI workflow
CPU → TX FIFO → SPI Core → RX FIFO → CPU

Loader Mode:

Send dummy byte to generate SCK
Converts every valid MISO byte into a one-cycle SRAM byte-write



SPI core

- Shift register
- Master FSM
- CPOL/CPHA logic
- Baud-rate generator
- TX/RX buffers

Serial pin drivers

- MOSI: drive data out
- MISO: sample data in
- SCK: generated clock
- NSS: chip select

SPI Register Model

Register	Purpose	Notes
SPI_CR1	Enable, master mode, CPOL, CPHA, bit order, divider	Main configuration register
SPI_CR2	Register for expansion	Reserved for interrupts/DMA
SPI_SR	Status flags: TXE, RXNE, BSY	Software observes buffer state and active transfer
SPI_DR	Transmit / receive data register	Write to transmit path; read received data

Status flags

TXE Transmit buffer available

RXNE Receive data available

BSY Active serial transaction

Logic Synthesis Flow

Physical Implementation

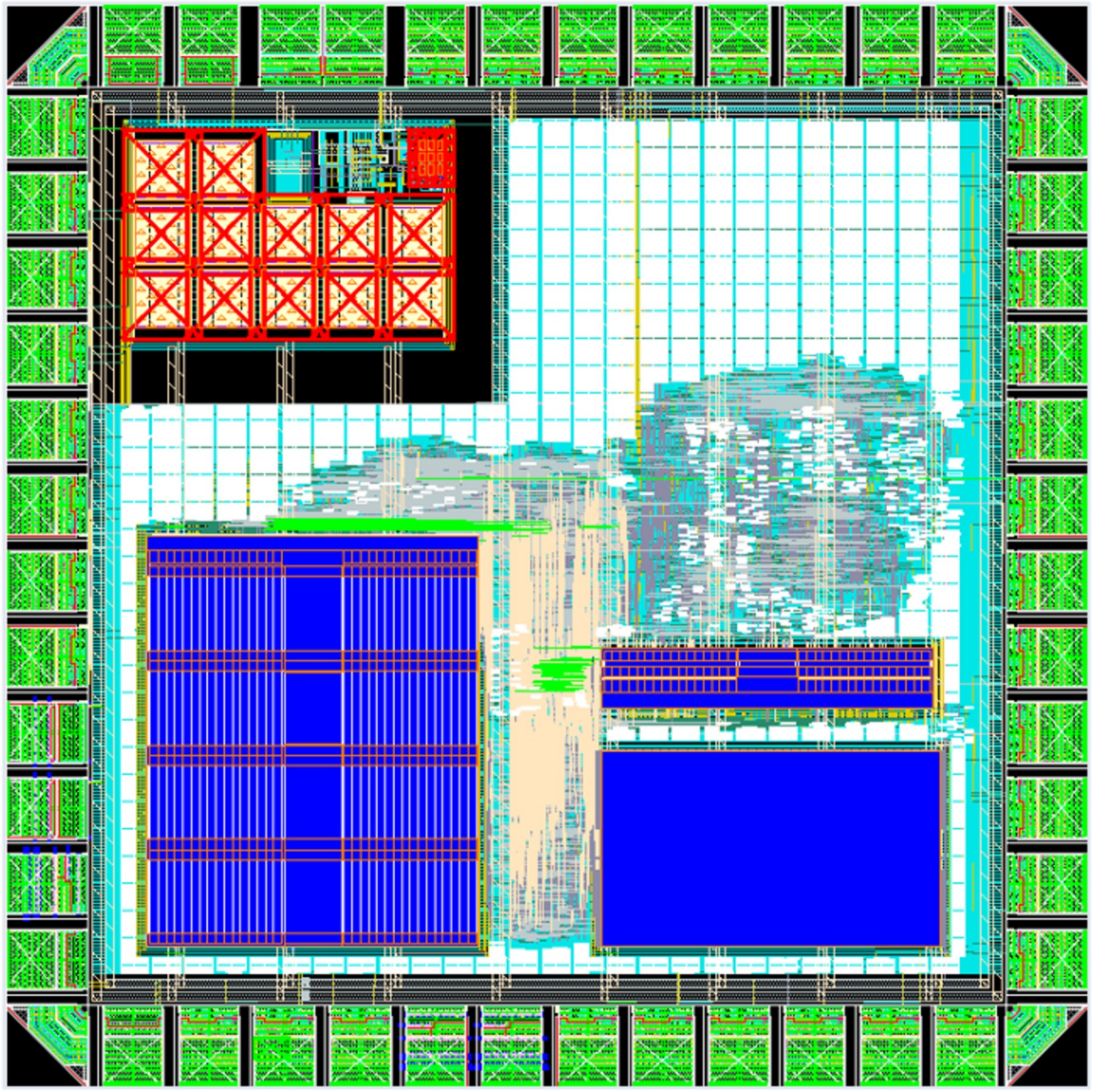
Analog / digital isolation

CPU to SRAM Floorplanning

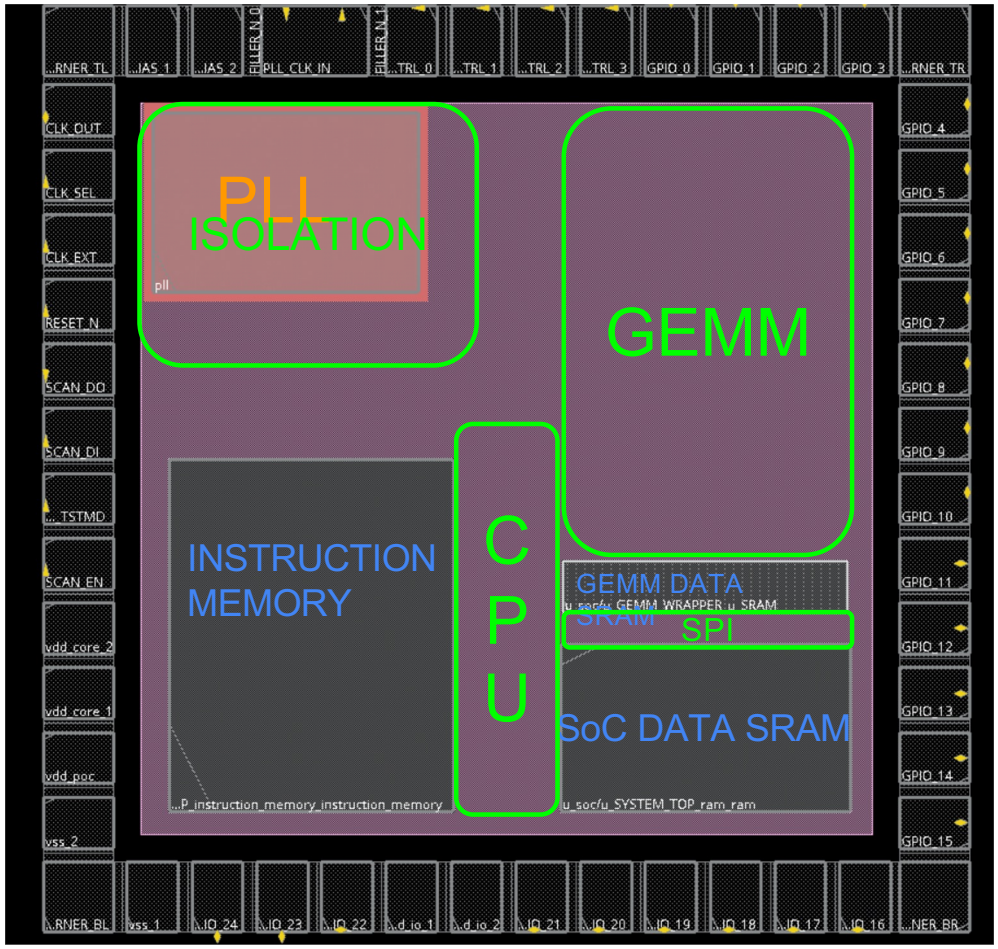
Dedicated clock-tree corridor for cleaner CTS and lower skew

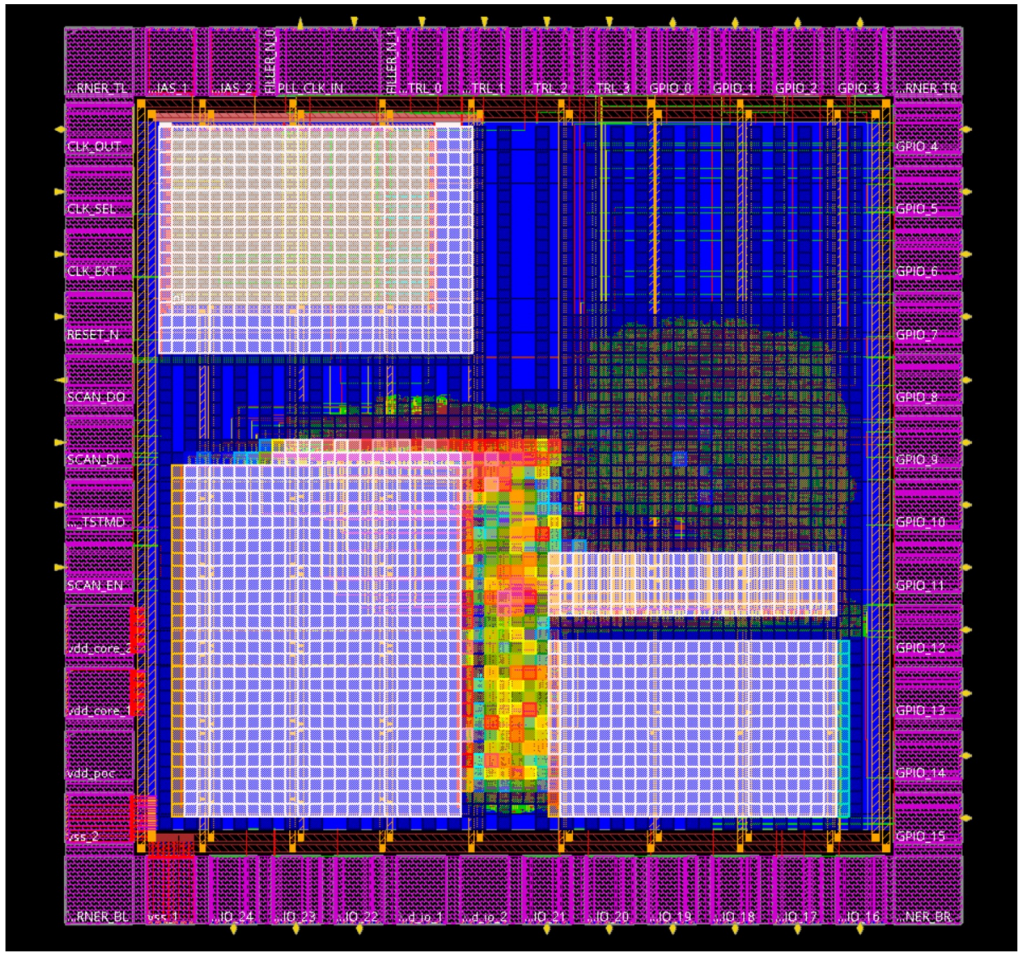
Macro placement to reduce routing congestion

Power integrity and routability awareness

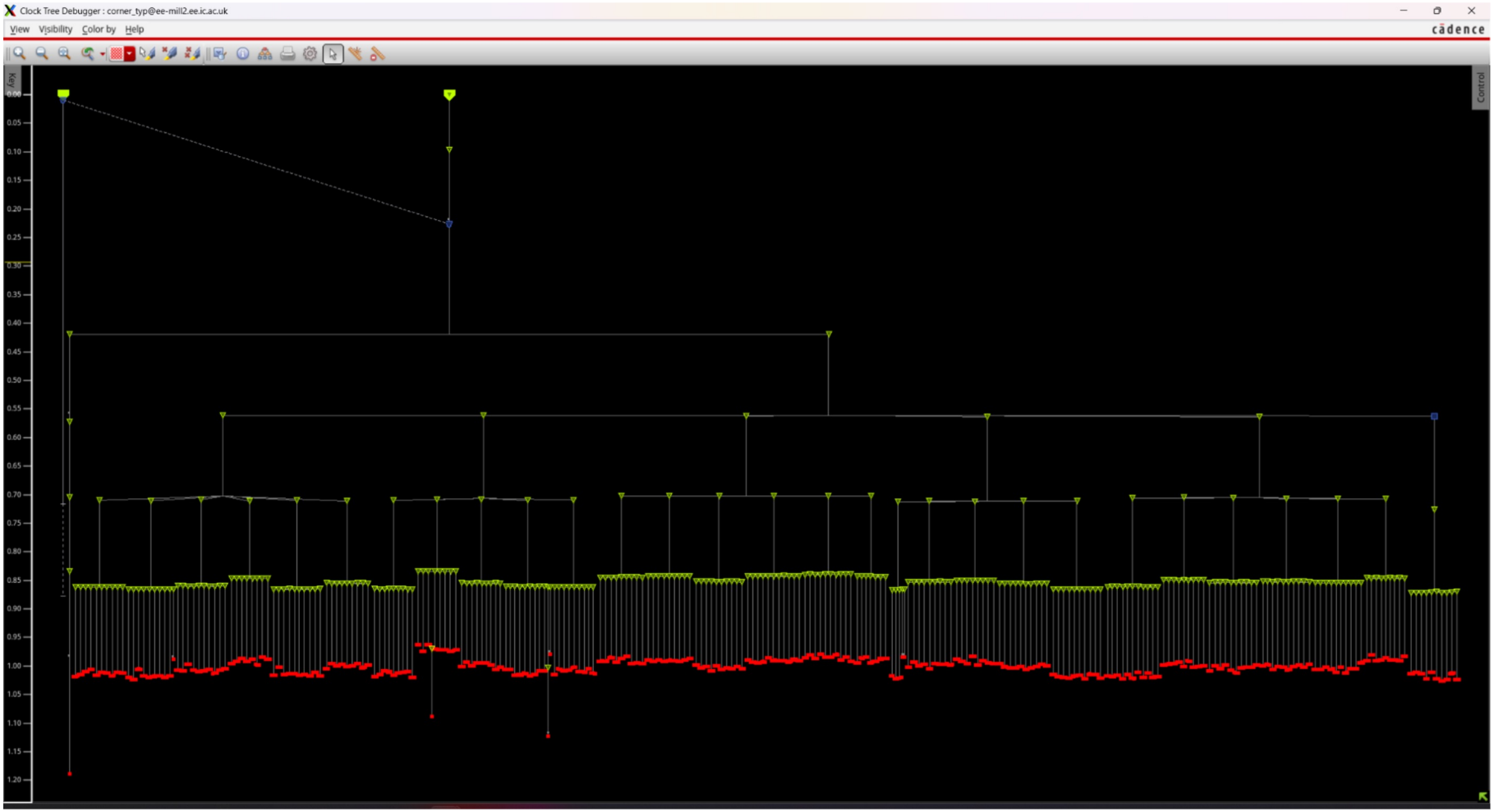


Macro Placement





CTS


Clock Tree Debugger : corner_typ@ee-mill2.ee.ic.ac.uk
View Visibility Color by Help
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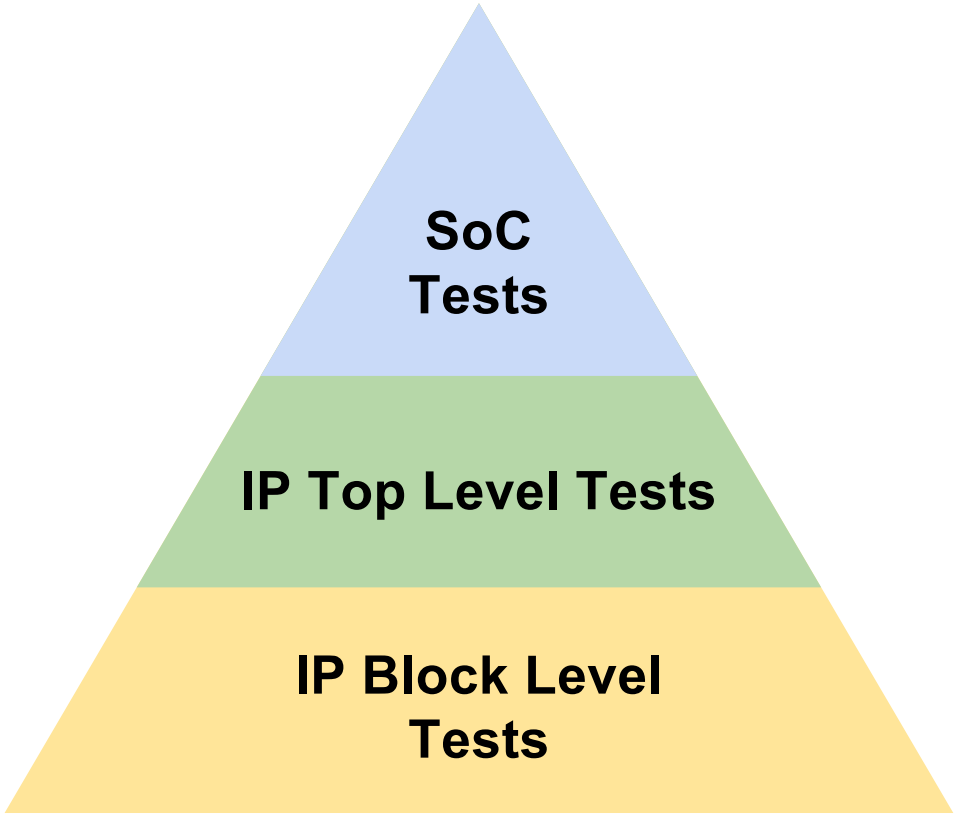
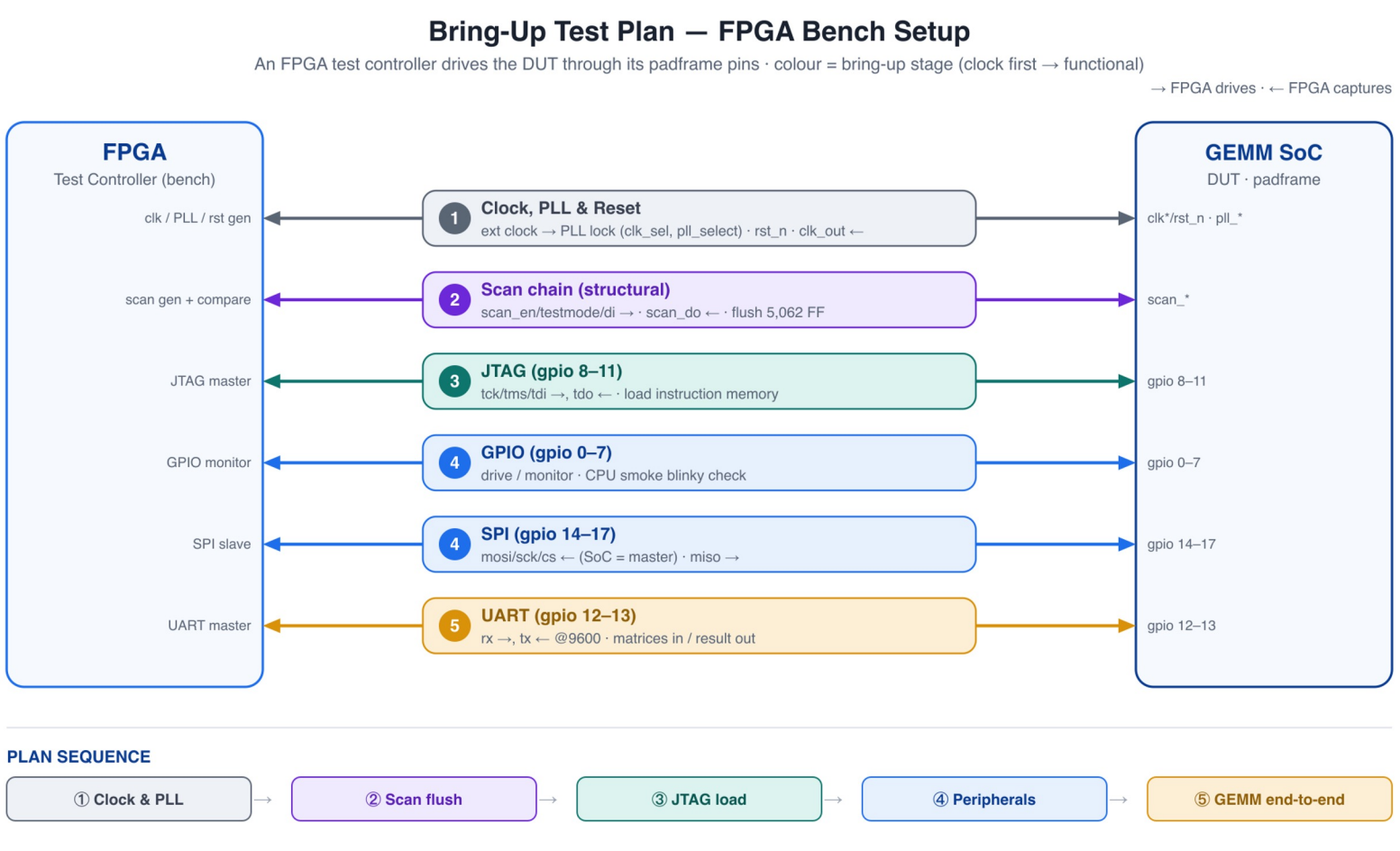
Verification

- 3 tier verification
- All IP components verified independently
- Cocotb + external libraries to drive Wishbone, JTAG and UART interfaces
- ~90 unique test cases, all passing
- Random overnight regression testing



Bring-Up Test Plan



Reflections

Possible improvements:

- Architectural improvements:
  - By using more Systolic Arrays and adder-trees, we can achieve larger-scale matrix multiplication acceleration.
  - By adding vector-related units, we would be able to support Flash Attention.
- Verification improvements:
  - Coverage metrics
  - Formal verification
- Time management